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Blockchain malware protection

Abstract

A request to add a new block to a blockchain is received. Data associated with the new block is scanned to identify malware and/or an anomaly. In response to identifying the malware and/or the anomaly in the data associated with the new block, an action is taken. The action includes: rejecting the request to add the new block to the blockchain, or removing the malware/anomaly from the new block and adding the new block to the blockchain. In a second embodiment, a malware event is identified that identifies malware/an anomaly in a block in a blockchain. In response to the malware event, an action is taken. The action includes: consolidating the blockchain, bypassing the block in the blockchain, consolidating the blockchain and bypassing the block in the blockchain, and deleting an encryption key that was used to encrypt the associated data that comprises the malware and/or the anomaly.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. patent application Ser. No. 17/356,164, filed Jun. 23, 2021. Cross reference is made to Patent Cooperation Treaty (PCT) Application Serial No. PCT/US2021/033207, filed May 19, 2021, entitled "Blockchain Consolidation with Active Archiving", which is incorporated herein by this reference in its entirety.

FIELD

(1) The disclosure relates generally to blockchains and particularly to protection of blockchains from malware and/or anomalies.

BACKGROUND

(2) One of the key advantages with blockchains is that the data stored/pointed to by the blockchain is immutable. However, this may cause a problem if a blockchain has been compromised by malware. For example, if a block in a blockchain contains malware that is activated when someone accesses the data in the block, this can cause problems anytime someone accesses the block that contains the malware.

(3) In addition, a copy of malware may be stored by the blockchain or pointed to by the blockchain. For example, a malware application that does not actually contain the malware may use a block in the blockchain as a reference to propagate a virus stored in the blockchain to other locations on the network. The problem is, once a block in the blockchain has been inserted or become infected with malware, how can the malware in the blockchain be mitigated without compromising the immutability of the blockchain?

SUMMARY

(4) These and other needs are addressed by the various embodiments and configurations of the present disclosure. The present disclosure can provide a number of advantages depending on the particular configuration. These and other advantages will be apparent from the disclosure contained herein.

(5) A request to add a new block to a blockchain is received. Data associated with the new block is scanned to identify malware and/or an anomaly. In response to identifying the malware and/or the anomaly in the data associated with the new block, an action is taken. The action includes: rejecting the request to add the new block to the blockchain, or removing the malware/anomaly from the new block and adding the new block to the blockchain.

(6) In a second embodiment, a malware event is identified that identifies malware/an anomaly in a block in a blockchain. In response to the malware event, an action is taken. The action includes: consolidating the blockchain, bypassing the block in the blockchain, consolidating the blockchain and bypassing the block in the blockchain, and deleting additional associated controls or data elements (such as encryption key used to encrypt the associated data) which comprise the malware and/or the anomaly.

(7) The phrases "at least one", "one or more", "or", and "and/or" are open-ended expressions that

are both conjunctive and disjunctive in operation. For example, each of the expressions “at least one of A, B and C”, “at least one of A, B, or C”, “one or more of A, B, and C”, “one or more of A, B, or C”, “A, B, and/or C”, and “A, B, or C” means A alone, B alone, C alone, A and B together, A and C together, B and C together, or A, B and C together.

(8) The term “a” or “an” entity refers to one or more of that entity. As such, the terms “a” (or “an”), “one or more” and “at least one” can be used interchangeably herein. It is also to be noted that the terms “comprising”, “including”, and “having” can be used interchangeably.

(9) The term “automatic” and variations thereof, as used herein, refers to any process or operation, which is typically continuous or semi-continuous, done without material human input when the process or operation is performed. However, a process or operation can be automatic, even though performance of the process or operation uses material or immaterial human input, if the input is received before performance of the process or operation. Human input is deemed to be material if such input influences how the process or operation will be performed. Human input that consents to the performance of the process or operation is not deemed to be “material”.

(10) Aspects of the present disclosure may take the form of an entirely hardware embodiment, an entirely software embodiment (including firmware, resident software, micro-code, etc.) or an embodiment combining software and hardware aspects that may all generally be referred to herein as a “circuit,” “module” or “system.” Any combination of one or more computer readable medium(s) may be utilized. The computer readable medium may be a computer readable signal medium or a computer readable storage medium.

(11) A computer readable storage medium may be, for example, but not limited to, an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system, apparatus, or device, or any suitable combination of the foregoing. More specific examples (a non-exhaustive list) of the computer readable storage medium would include the following: an electrical connection having one or more wires, a portable computer diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), an optical fiber, a portable compact disc read-only memory (CD-ROM), an optical storage device, a magnetic storage device, or any suitable combination of the foregoing. In the context of this document, a computer readable storage medium may be any tangible medium that can contain, or store a program for use by or in connection with an instruction execution system, apparatus, or device.

(12) A computer readable signal medium may include a propagated data signal with computer readable program code embodied therein, for example, in baseband or as part of a carrier wave. Such a propagated signal may take any of a variety of forms, including, but not limited to, electromagnetic, optical, or any suitable combination thereof. A computer readable signal medium may be any computer readable medium that is not a computer readable storage medium and that can communicate, propagate, or transport a program for use by or in connection with an instruction execution system, apparatus, or device. Program code embodied on a computer readable medium may be transmitted using any appropriate medium, including but not limited to wireless, wireline, optical fiber cable, RF, etc., or any suitable combination of the foregoing.

(13) The terms “determine”, “calculate” and “compute,” and variations thereof, as used herein, are used interchangeably and include any type of methodology, process, mathematical operation or technique.

(14) The term “means” as used herein shall be given its broadest possible interpretation in accordance with 35 U.S.C., Section 112(f) and/or Section 112, Paragraph 6. Accordingly, a claim incorporating the term “means” shall cover all structures, materials, or acts set forth herein, and all of the equivalents thereof. Further, the structures, materials or acts and the equivalents thereof shall include all those described in the summary, brief description of the drawings, detailed description, abstract, and claims themselves.

(15) The term “blockchain” as described herein and in the claims refers to a growing list of records,

called blocks, which are linked using cryptography. The blockchain is commonly a decentralized, distributed and public digital ledger that is used to record transactions across many computers so that the record cannot be altered retroactively without the alteration of all subsequent blocks and the consensus of the network. Each block contains a cryptographic hash of the previous block, a timestamp, and transaction data (generally represented as a merkle tree root hash). For use as a distributed ledger, a blockchain is typically managed by a peer-to-peer network collectively adhering to a protocol for inter-node communication and validating new blocks. Once recorded, the data in any given block cannot be altered retroactively without alteration of all subsequent blocks, which requires consensus of the network majority. In verifying or validating a block in the blockchain, a hashcash algorithm generally requires the following parameters: a service string, a nonce, and a counter. The service string can be encoded in the block header data structure, and include a version field, the hash of the previous block, the root hash of the merkle tree of all transactions (or information or data) in the block, the current time, and the difficulty level. The nonce can be stored in an extraNonce field, which is stored as the left most leaf node in the merkle tree. The counter parameter is often small at 32-bits so each time it wraps the extraNonce field must be incremented (or otherwise changed) to avoid repeating work. When validating or verifying a block, the hashcash algorithm repeatedly hashes the block header while incrementing the counter & extraNonce fields. Incrementing the extraNonce field entails recomputing the merkle tree, as the transaction or other information is the left most leaf node. The body of the block contains the transactions or other information. These are hashed only indirectly through the Merkle root.

(16) As described herein and in the claims, the term “malware” can be any malicious software, such as, a virus, a keylogger, a worm, a trojan, ransomware/crypto-malware, logic bombs, bots/botnets, addware, spyware, rootkits, intentionally malformed data, and/or the like. Malware can infect any portion of a block in a blockchain, such as data in the block, data pointed by the block, and/or the like.

(17) As described herein and in the claims, the term “malware event” may be any event related to the detection of a malicious software, such as the detection of malware, detection of an anomaly, detection of unusual behaviors, detection of unusual traffic patterns, detection of unusual user patterns, detection of different thread activity, detection of different access patterns, and/or the like.

(18) The preceding is a simplified summary to provide an understanding of some aspects of the disclosure. This summary is neither an extensive nor exhaustive overview of the disclosure and its various embodiments. It is intended neither to identify key or critical elements of the disclosure nor to delineate the scope of the disclosure but to present selected concepts of the disclosure in a simplified form as an introduction to the more detailed description presented below. As will be appreciated, other embodiments of the disclosure are possible utilizing, alone or in combination, one or more of the features set forth above or described in detail below. Also, while the disclosure is presented in terms of exemplary embodiments, it should be appreciated that individual aspects of the disclosure can be separately claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block diagram of a first illustrative system that shows a blockchain in a distributed ledger.

(2) FIG. 2 is a block diagram of a process for identifying malware being added to a blockchain.

(3) FIG. 3 is a block diagram of a process for consolidating a blockchain based on a malware event.

(4) FIG. 4 is a block diagram of a process for consolidating a blockchain based on a malware event by bypassing an infected block.

(5) FIG. 5 is a block diagram of a process for removing access to a malware/anomaly block in a

blockchain.

(6) FIG. 6 is a flow diagram of a process for identifying malware/anomalies being added to a blockchain.

(7) FIG. 7 is a flow diagram of a process for managing malware in a new block that has been requested to be added to a blockchain.

(8) FIG. 8 is a flow diagram of a process for mitigating malware that has been stored in a blockchain.

DETAILED DESCRIPTION

(9) FIG. 1 is a block diagram of a first illustrative system **100** that shows a blockchain **102** in a distributed blockchain ledger **120**. The first illustrative system **100** comprises nodes **101A-101N**, a communication device **105**, and a network **110**.

(10) The nodes **101A-101N** can be or may include any hardware/software that can support the use of blockchains **102** in the distributed blockchain ledger **120**, such as, a Personal Computer (PC), a server, a trust authority server, a gateway, a router, a network device, and/or the like. As shown in FIG. 1, any number of nodes **101A-101N** may be connected to the network **110**. The nodes **101A-101N** further comprise blockchains **102A-102N**, blockchain managers **103A-103N**, and malware/anomaly detectors **104A-104N**.

(11) The blockchains **102A-102N** are copies of the same blockchain **102** that comprise the distributed blockchain ledger **120**. The size of the blockchains **102A-102N** may vary based on implementation. The blockchains **102A-102N** are a form of a replicated distributed database.

(12) The blockchain managers **103A-103N** can be or may include any hardware coupled with software that can manage the blockchains **102A-102N**. The blockchain managers **103A-103N** work together to determine how to manage the blockchains **102A-102N**. For example, the blockchain managers **103A-103N** may vote to validate a new block being added to the blockchains **102A-102N** in the distributed blockchain ledger **120**.

(13) The malware/anomaly detectors **104A-104N** can be or may include any software coupled with software that detect various kinds of malware/anomalies. The malware/anomaly detectors **104A-104N** may work in conjunction to identify infected blocks that are either already in the blockchains **102A-102N** or new blocks that are being added to the blockchains **102A-102N**.

(14) The communication device **105** can be or may include any device that can communicate on the network **110**, such as a Personal Computer (PC), a cellular telephone, a Personal Digital Assistant (PDA), a tablet device, a notebook device, a smartphone, and/or the like. The communication device **105** can be used to administer the blockchain managers **103A-103N** and/or the malware/anomaly detectors **104A-104N**.

(15) The network **110** can be or may include any collection of communication equipment that can send and receive electronic communications, such as the Internet, a Wide Area Network (WAN), a Local Area Network (LAN), a packet switched network, a circuit switched network, a cellular network, a combination of these, and the like. The network **110** can use a variety of electronic protocols, such as Ethernet, Internet Protocol (IP), Hyper Text Markup Language (HTML), Hyper Text Transfer Protocol (HTTP), Web Real-Time Protocol (Web RTC), and/or the like. Thus, the network **110** is an electronic communication network configured to carry messages via packets and/or circuit switched communications.

(16) FIG. 2 is a block diagram of a process for identifying malware being added to a blockchain **102**. FIG. 2 is an example of an exemplary node **101** that comprises the blockchain **102**, the blockchain manager **103**, and the malware/anomaly detector **104**. Each node **101** in the distributed ledger **120** may work in the same manner as described in FIG. 2. In FIG. 2, the blockchain **102** comprises a genesis block **200A** and block **200N** (where N means that any number of blocks may be in the blockchain **102**). The block **200N** has a forward hash of the genesis block **200A**. The block **200N** has a forward link **201A** to the genesis block **200A**.

(17) When a new block **200N+1** has been requested to be added to the blockchain **102** in step **210**,

the new block **200N+1** is received by the blockchain manager **103**. The blockchain manager **103** requests the malware/anomaly detector **104** to scan the new block **200N+1** in step **211**. The new block **200N+1** is scanned by a malware/anomaly detector **104** in step **212**. The malware/anomaly detector **104** can scan for known virus/malware patterns, anomalies compared to previous blocks **200A-200N**/blockchains **102**, and/or the like. The malware patterns could come from a centralized point. The anomaly data could be data that is learned based on machine learning (e.g., unsupervised machine learning). The scanning may include active code that checks for lengths of data so that a buffer overflow condition is not created where malware code can be executed when accessing a block in the blockchain **102** (similar to how emails/PDFs, Word documents have been compromised). This allows for detection of malware that loads based on accessing the data of the blockchain. Once the scan is complete, the malware/anomaly detector **104** notifies the blockchain manager **103** if the new block **200N+1** contains a malware/anomaly in step **213**. If the new block **200N+1** does not contain and malware/anomaly, the new block **200N+1** is added to the blockchain **102** in step **214**. If the new block **200N+1** does contain a malware/anomaly, the malware/anomaly detector **104** may remove the malware before the new block **200N+1** is added to the blockchain **102** in step **214**.

(18) Each node **101** in the distributed blockchain ledger **120** may have a malware/anomaly detector **104A-104N** that actively checks for malware being added to any new block **200N+1** in the blockchain **102**. By having each node **101** in the distributed blockchain ledger **120** do this, when the new block **200N+1** is requested to be added, the nodes **101A-101N** in the distributed blockchain ledger **120** have to have a consensus that the new block **200N+1** does not contain any malware/anomaly before the new block **200N+1** is added to the distributed blockchain ledger **120**. Basically anytime a new block **200N+1** is added, the new block **200N+1** is actively checked for malicious code or indicators that may indicate that malicious code/data is being added to the blockchain **102** (either directly in the blockchain **102** or pointed to by the blockchain **102**).

(19) FIG. 3 is a block diagram of a process for consolidating a blockchain **102** based on a malware event **310**. FIG. 3 comprises the blockchain **102** and a consolidation blockchain **300**. The blockchain **102** comprises the genesis block **200A**, block **200B**, malware/anomaly block **200C**, and block **200N**. The malware/anomaly block **200C** contains or links to malware/anomaly data. The blocks **200A-200N** are linked together via links **201A-201C** as is traditionally done in blockchains **102**.

(20) The consolidation blockchain **300** comprises a consolidation genesis block **200AC** and block **200NC**. The consolidation blockchain **300** is linked via link **201E** similar to the blockchain **102**.

(21) After creation of the blockchain **102** (including blocks **200A-200N**), some type of malware/anomaly is detected in the blockchain **102** (a malware event **310**). The detection of the malware/anomaly can be determined in various ways, such as where the malware/anomaly detector **104** receives a new virus pattern and detects that the malware/anomaly block **200C** has been infected with the virus or has a pointer that points data that includes the virus. Based on the malware event in step **310**, the blockchain manager **103** ends the blockchain **102**. Ending the blockchain **102** may comprise adding an end block **200M** to the blockchain **102** that identifies that block **200C** contains a malware/anomaly. The end block **200M** may comprise additional information about the malware/anomaly, such as, the type of malware/anomaly, the location of the malware/anomaly, and/or the like. The end block **200M** has a link **201D** that links to the block **200N**. The end block **200M** may also point to the consolidation blockchain **300**.

(22) The blockchain manager **103** then creates the consolidation blockchain **300**. The creation of the consolidation blockchain **300** comprises creating the consolidation genesis block **200AC**. The consolidation genesis block **200AC** contains a link **301** back to the blockchain **102**. In addition, the consolidation genesis block **200AC** contains information that identifies that the malware/anomaly block **200C** contains or points to malware. The information may indicate the type of malware and whether the malware is in the malware/anomaly block **200C** and/or pointed to by the

malware/anomaly block **200C**. The information may indicate that block is invalid. If it is later determined that that there is not actually malware in the block **200C** (or pointed to by block **200C**), a block may be added to the consolidation blockchain that the block **200C** is actually valid. The consolidation genesis block **200AC** also comprises a hash of the blockchain **102**. This is to prevent someone changing or replacing the blockchain **102**.

(23) After creation of the consolidation genesis block **200AC**, blocks (e.g., block **200NC**) are added in the traditional manner to the consolidation blockchain **300**. In this illustrative embodiment, block **200NC** has been added to the consolidation blockchain **300**. The block **200NC** has a forward hash and a link **201E** to the consolidation genesis block **200AC** (similar to the blockchain **102**).

(24) By creating the consolidation blockchain **300**, the malware/anomaly of malware/anomaly block **200C** can effectively be bypassed because of the creation of the consolidation blockchain **300**. If data in the blockchain **102** needs to be accessed, an application/user who wants to access the blockchain **102** is made aware that the blockchain **102** contains malware in malware/anomaly block **200C** or points to data with malware/anomalies by looking at the information in the consolidation genesis block **200AC**.

(25) FIG. 4 is a block diagram of a process for consolidating a blockchain **102** based on a malware event **310** by bypassing the malware/anomaly block **200C**. The process of creating the consolidation blockchain **300** based on the malware event **310** works the same as described above in FIG. 3. The difference between FIG. 3 and FIG. 4 is that the data in the consolidation genesis block **200AC** also indicates that the malware/anomaly block **200C** has been bypassed.

(26) The malware/anomaly block **200C** is bypassed by creating bypass block **400** that has the malware removed (or is removed from the data pointed to by the malware/anomaly block **200C**). A new forward hash is created that has links **401A-401C** that links the blocks **200N**, **400**, **200B**, and **200A** together. Thus, when an application/user want to access the blockchain **102**, by looking at the consolidation genesis block **200AC**, the application/user will know to use the bypass block **400** instead of the malware/anomaly block **200C**. The new forward hashes that are linked together via links **401A-401C** can be used to validate the bypass block **400**.

(27) In another embodiment, instead of creating a consolidation blockchain **300**, the malware/anomaly block **200C** is just bypassed. When the malware/anomaly is detected, a malware/anomaly ID block **402** is added to the blockchain **102** that identifies the malware/anomaly block **200C** and indicates that the malware/anomaly block **200C** has been bypassed (like described above). The malware/anomaly ID block **402** has a hash of the bypass block **400** to prevent a substitute for the bypass block **400**. The malware/anomaly ID block **402** also has a link **401D** that links to the block **200N**. The malware/anomaly ID block **402** has a hash of the block **200N**. After the malware/anomaly ID block **402** has been added, additional blocks can then be added to the end of the blockchain **102**.

(28) FIG. 5 is a block diagram of a process for removing access to a malware/anomaly block **200C** in a blockchain **102**. FIG. 5 comprises the blockchain **102**. The blockchain **102** further comprises the blocks **200A-200N** with the links **201A-201C**. In FIG. 5, the blocks **200B-200N** has pointers to data **500B-500N**. Although not shown, the data **500B-500N** may actually reside within the blocks **200B-200N**.

(29) When a new block is added to the blockchain **102**, the data associated with the new block is encrypted with an individual encryption key. For example, as shown in FIG. 5, the data **500B** is encrypted with key **501B**, the data **500C** is encrypted with key **501C**, and the data **500N** is encrypted with key **501N**. When an application/user wants to access the data **500B-500N**, the individual encryption keys **501B-501N** are used respectively to unencrypt the data **500B-500N**.

(30) When anomalies/malware is detected in the malware/anomaly block **200C**, the blockchain manager **103** deletes the encryption key **501C**. Thus access to the data **500C** can no longer occur. This prevents someone from accessing the data **500C**, which contains the malware/anomaly.

(31) FIG. 6 is a flow diagram of a process for identifying malware/anomalies being added to a

blockchain **102**. Illustratively, the nodes **101A-101N**, the blockchains **102A-102N**, the blockchain managers **103A-103N**, the malware/anomaly detectors **104A-104N**, and the communication device **105** are stored-program-controlled entities, such as a computer or microprocessor, which performs the method of FIGS. **6-8** and the processes described herein by executing program instructions stored in a computer readable storage medium, such as a memory (i.e., a computer memory, a hard disk, and/or the like). Although the methods described in FIGS. **6-8** are shown in a specific order, one of skill in the art would recognize that the steps in FIGS. **6-8** may be implemented in different orders and/or be implemented in a multi-threaded environment. Moreover, various steps may be omitted or added based on implementation.

(32) The process starts in step **600**. The blockchain manager **103** waits, in step **602**, for a request to add the new block **200N+1** to the blockchain **102**. If a request to add the new block **200N+1** to the blockchain **102** is not received in step **602**, the process of step **602** repeats. Otherwise, if a request to add the new block **200N+1** to the blockchain **102** is received in step **602**, the malware/anomaly detector **104** scans the data **500** associated with the new block **200N+1** to identify any malware/anomalies in step **604**.

(33) If there are no malware/anomalies detected in the new block **200N+1** in step **606**, the blockchain manager **103** adds, in step **610**, the new block **200N+1** to the blockchain **102**. The adding of the new block **200N+1** in step **610** is based on a consensus vote of the blockchain managers **103A-103N** in the distributed blockchain ledger **120**. The process then goes to step **612**. If there is malware detected in step **606**, the malware/anomaly detector **104** manages the request to add the new block **200N+1** in step **608**. For example, the malware/anomaly may be removed from the blockchain **102**. Step **608** is described in further detail in FIG. **7**. The process then goes to step **612**.

(34) The blockchain manager **103** determines, in step **612**, if the process is complete. If the process is not complete in step **612**, the process goes back to step **602**. Otherwise, if the process is complete in step **612**, the process ends in step **614**.

(35) FIG. **7** is a flow diagram of a process for managing malware/anomalies in a new block **200N+1** that has been requested to be added to the blockchain **102**. The process of FIG. **7** is an exemplary embodiment of step **608** of FIG. **6**. If malware is detected in the data in step **606**, the malware/anomaly detector **104** determines whether to remove the malware/anomaly from the block in step **700**. Step **700** may work in various ways. For example, the decision of step **700** may be administered, may be based on the type of malware/anomaly, based on a security level and/or type of the blockchain **102**, based on a user/application requesting to add the new block **200N+1** to the blockchain **102**, based on machine learning, and/or the like.

(36) If the malware/anomaly is not to be removed from the blockchain **102** in step **700**, the blockchain manager **103** rejects the request to add the new block **200N+1** to the blockchain **102** in step **706**. The process then goes to step **708**.

(37) Otherwise, if the decision is to remove the malware/anomaly in step **700**, the malware/anomaly detector **104** removes the malware/anomaly from the data in the new block **200N+1** or the data pointed to by the new block **200N+1** in step **702**. The blockchain manager **103** adds the new block **200N+1** to the blockchain **102** in step **704**. The blockchain manager **103** then sends a notification message of the rejected new block **200N+1** or the removal of the malware/anomaly in step **708**. For example, an administrator of the blockchain **102** at the communication device **105** may be notified in step **708**. The process then goes to step **612**.

(38) The process of steps **702**, **704**, and **706** are completed by the blockchain managers **103A-103N** based on a consensus vote of the blockchain managers **103A-103N**. A consensus vote is based on more than 50% agreement of the blockchain managers **103A-103N** to take the actions of steps **702**, **704**, and **706**.

(39) FIG. **8** is a flow diagram of a process for mitigating malware that has been stored in a blockchain **102**. The process starts in step **800**. The malware/anomaly detector **104**, in step **802**,

determines if a malware event has been identified in the blockchain **102**. A malware event may be based on various types of events, such as, detection of a virus in the blockchain **102**, detection of anomalous data in the blockchain **102**, detection of an malicious actor that added a block to the blockchain **102**, detection of a malicious application/user accessing a block(s) in the blockchain **102**, periodic access of the blockchain **102** where this previously never occurred (e.g., a malicious application is loading a virus out of the blockchain **102**), detection of a virus pattern in the blockchain **102**, detection of a malformed block in the blockchain **102**, detection of malicious behavior based on data **500** in the blockchain **102**, and/or the like. The detection of a malformed block may include detection of where the block (e.g., block **200C**) is intentionally malformed. In this case, the attack would be to either try and crash the block chain **121** processing or generate a buffer overflow. In the case of the buffer overflow, the attack would cause the validation process to do something bad or the compromise the process.

(40) If a malware event is not identified in step **802**, the process of step **802** repeats. Otherwise, if the malware event is identified in step **802**, the malware/anomaly detector **104** determines the type of action to take. In FIG. **8**, the malware/anomaly detector **104** may take different actions. The first option is to consolidate the blockchain **102** with using the bypass in step **806** as described above in FIG. **4**. Here the blockchain manager **103** creates the consolidation genesis block **200AC** (that indicates that there is malware/an anomaly in the malware/anomaly block **200C**), the bypass block **400**, and the new hashes/links **401A-401C** like shown and described in FIG. **4**. The process then goes to step **814**.

(41) Alternatively, the action may be to just consolidate the blockchain **102** in step **808** as described in above in FIG. **3**. In step **808**, the consolidation genesis block **200AC** (the start of the consolidation blockchain **300**) is created and optionally the end block **200M**. The consolidation genesis block **200AC** indicates that the malware/anomaly block **200C** contains malware and/or one or more anomalies. The information in the consolidation genesis block **200AC** may include the type of malware, information on how the malware works, whether the malware is located in the malware/anomaly block **200C** or is pointed to by the malware/anomaly block **200C**, if there is an anomaly in the malware/anomaly block **200C**, if there is an anomaly pointed to by the malware/anomaly block **200C**, the type of anomaly, who recommended adding the malware block **200AC**, any other type of information related to the malware/anomaly, and/or the like. The process then goes to step **814**.

(42) Another option is to bypass the malware/anomaly block **200C** in step **810**. Like described in FIG. **4**, the malware/anomaly ID block **402** is added to the blockchain **102** that identifies the malware/anomaly block **200C** and indicates that the malware/anomaly block **200C** has been bypassed. The bypass block **400** is added along with the links **401A-401C** that link the bypass block **400** to the blockchain **102**.

(43) In step **810**, the process may bypass the malware/anomaly block **200C** by deleting a pointer to the data (e.g., the pointer in block **200C** that points to the data **500C**) that contains the malware. The process then goes to step **814**.

(44) Alternatively, the action may be to delete the individual encryption key (e.g., encryption key **501C**) that is associated with the malware/anomaly block **200C** in step **812** as described above in FIG. **5**. This way, the malware/anomaly data is encrypted and cannot be accessed. Thus, the malware/anomaly is now benign. The process then goes to step **814**.

(45) In some embodiments, the malware/anomaly detector **104** may only have two or even one option in step **804**. The type of action that is taken in step **804** may be administered based on the type of malware event and/or anomaly. The action taken in step **804** may be based on the configuration of the blockchain **102**. For example, if the blockchain **102** was created using the encryption key method of FIG. **5**, then the process would delete the encryption key in step **812**.

(46) The blockchain manager **103** determines, in step **814** if the process is complete. If the process is not complete in step **814**, the process goes back to step **802**. Otherwise, if the process is complete

in step **814**, the process ends in step **816**.

(47) Examples of the processors as described herein may include, but are not limited to, at least one of Qualcomm® Snapdragon® 800 and 801, Qualcomm® Snapdragon® 610 and 615 with 4G LTE Integration and 64-bit computing, Apple® A7 processor with 64-bit architecture, Apple® M7 motion coprocessors, Samsung® Exynos® series, the Intel® Core™ family of processors, the Intel® Xeon® family of processors, the Intel® Atom™ family of processors, the Intel Itanium® family of processors, Intel® Core® i5-4670K and i7-4770K 22 nm Haswell, Intel® Core® i5-3570K 22 nm Ivy Bridge, the AMD® FX™ family of processors, AMD® FX-4300, FX-6300, and FX-8350 32 nm Vishera, AMD® Kaveri processors, Texas Instruments® Jacinto C6000™ automotive infotainment processors, Texas Instruments® OMAP™ automotive-grade mobile processors, ARM® Cortex™-M processors, ARM® Cortex-A and ARM1926EJ-S™ processors, other industry-equivalent processors, and may perform computational functions using any known or future-developed standard, instruction set, libraries, and/or architecture.

(48) Any of the steps, functions, and operations discussed herein can be performed continuously and automatically.

(49) However, to avoid unnecessarily obscuring the present disclosure, the preceding description omits a number of known structures and devices. This omission is not to be construed as a limitation of the scope of the claimed disclosure. Specific details are set forth to provide an understanding of the present disclosure. It should however be appreciated that the present disclosure may be practiced in a variety of ways beyond the specific detail set forth herein.

(50) Furthermore, while the exemplary embodiments illustrated herein show the various components of the system collocated, certain components of the system can be located remotely, at distant portions of a distributed network, such as a LAN and/or the Internet, or within a dedicated system. Thus, it should be appreciated, that the components of the system can be combined in to one or more devices or collocated on a particular node of a distributed network, such as an analog and/or digital telecommunications network, a packet-switch network, or a circuit-switched network. It will be appreciated from the preceding description, and for reasons of computational efficiency, that the components of the system can be arranged at any location within a distributed network of components without affecting the operation of the system. For example, the various components can be located in a switch such as a PBX and media server, gateway, in one or more communications devices, at one or more users' premises, or some combination thereof. Similarly, one or more functional portions of the system could be distributed between a telecommunications device(s) and an associated computing device.

(51) Furthermore, it should be appreciated that the various links connecting the elements can be wired or wireless links, or any combination thereof, or any other known or later developed element(s) that is capable of supplying and/or communicating data to and from the connected elements. These wired or wireless links can also be secure links and may be capable of communicating encrypted information. Transmission media used as links, for example, can be any suitable carrier for electrical signals, including coaxial cables, copper wire and fiber optics, and may take the form of acoustic or light waves, such as those generated during radio-wave and infra-red data communications.

(52) Also, while the flowcharts have been discussed and illustrated in relation to a particular sequence of events, it should be appreciated that changes, additions, and omissions to this sequence can occur without materially affecting the operation of the disclosure.

(53) A number of variations and modifications of the disclosure can be used. It would be possible to provide for some features of the disclosure without providing others.

(54) In yet another embodiment, the systems and methods of this disclosure can be implemented in conjunction with a special purpose computer, a programmed microprocessor or microcontroller and peripheral integrated circuit element(s), an ASIC or other integrated circuit, a digital signal processor, a hard-wired electronic or logic circuit such as discrete element circuit, a programmable

logic device or gate array such as PLD, PLA, FPGA, PAL, special purpose computer, any comparable means, or the like. In general, any device(s) or means capable of implementing the methodology illustrated herein can be used to implement the various aspects of this disclosure. Exemplary hardware that can be used for the present disclosure includes computers, handheld devices, telephones (e.g., cellular, Internet enabled, digital, analog, hybrids, and others), and other hardware known in the art. Some of these devices include processors (e.g., a single or multiple microprocessors), memory, nonvolatile storage, input devices, and output devices. Furthermore, alternative software implementations including, but not limited to, distributed processing or component/object distributed processing, parallel processing, or virtual machine processing can also be constructed to implement the methods described herein.

(55) In yet another embodiment, the disclosed methods may be readily implemented in conjunction with software using object or object-oriented software development environments that provide portable source code that can be used on a variety of computer or workstation platforms.

Alternatively, the disclosed system may be implemented partially or fully in hardware using standard logic circuits or VLSI design. Whether software or hardware is used to implement the systems in accordance with this disclosure is dependent on the speed and/or efficiency requirements of the system, the particular function, and the particular software or hardware systems or microprocessor or microcomputer systems being utilized.

(56) In yet another embodiment, the disclosed methods may be partially implemented in software that can be stored on a storage medium, executed on programmed general-purpose computer with the cooperation of a controller and memory, a special purpose computer, a microprocessor, or the like. In these instances, the systems and methods of this disclosure can be implemented as program embedded on personal computer such as an applet, JAVA® or CGI script, as a resource residing on a server or computer workstation, as a routine embedded in a dedicated measurement system, system component, or the like. The system can also be implemented by physically incorporating the system and/or method into a software and/or hardware system.

(57) Although the present disclosure describes components and functions implemented in the embodiments with reference to particular standards and protocols, the disclosure is not limited to such standards and protocols. Other similar standards and protocols not mentioned herein are in existence and are considered to be included in the present disclosure. Moreover, the standards and protocols mentioned herein and other similar standards and protocols not mentioned herein are periodically superseded by faster or more effective equivalents having essentially the same functions. Such replacement standards and protocols having the same functions are considered equivalents included in the present disclosure.

(58) The present disclosure, in various embodiments, configurations, and aspects, includes components, methods, processes, systems and/or apparatus substantially as depicted and described herein, including various embodiments, subcombinations, and subsets thereof. Those of skill in the art will understand how to make and use the systems and methods disclosed herein after understanding the present disclosure. The present disclosure, in various embodiments, configurations, and aspects, includes providing devices and processes in the absence of items not depicted and/or described herein or in various embodiments, configurations, or aspects hereof, including in the absence of such items as may have been used in previous devices or processes, e.g., for improving performance, achieving ease and/or reducing cost of implementation.

(59) The foregoing discussion of the disclosure has been presented for purposes of illustration and description. The foregoing is not intended to limit the disclosure to the form or forms disclosed herein. In the foregoing Detailed Description for example, various features of the disclosure are grouped together in one or more embodiments, configurations, or aspects for the purpose of streamlining the disclosure. The features of the embodiments, configurations, or aspects of the disclosure may be combined in alternate embodiments, configurations, or aspects other than those discussed above. This method of disclosure is not to be interpreted as reflecting an intention that

the claimed disclosure requires more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive aspects lie in less than all features of a single foregoing disclosed embodiment, configuration, or aspect. Thus, the following claims are hereby incorporated into this Detailed Description, with each claim standing on its own as a separate preferred embodiment of the disclosure.

(60) Moreover, though the description of the disclosure has included description of one or more embodiments, configurations, or aspects and certain variations and modifications, other variations, combinations, and modifications are within the scope of the disclosure, e.g., as may be within the skill and knowledge of those in the art, after understanding the present disclosure. It is intended to obtain rights which include alternative embodiments, configurations, or aspects to the extent permitted, including alternate, interchangeable and/or equivalent structures, functions, ranges or steps to those claimed, whether or not such alternate, interchangeable and/or equivalent structures, functions, ranges or steps are disclosed herein, and without intending to publicly dedicate any patentable subject matter.

Claims

1. A system, comprising: a microprocessor; and a computer readable medium, coupled with the microprocessor and comprising microprocessor readable and executable instructions that, when executed by the microprocessor, cause the microprocessor to: identify a malware event, wherein the malware event identifies a block in a blockchain that has associated data that comprises malware and/or an anomaly; and in response to identifying the malware event that identifies the block in the blockchain that has the associated data that comprises the malware and/or the anomaly, doing at least one of the following actions: consolidate the blockchain; consolidate the blockchain and bypass the block in the blockchain that comprises the malware and/or anomaly; and delete an encryption key that was used to encrypt the associated data that comprises the malware and/or the anomaly.
2. The system of claim 1, wherein the malware event is based on at least one of the following: detection of a virus in the blockchain; detection of anomalous data in the blockchain; detection of a malicious application/user accessing one or more blocks in the blockchain; detection of abnormal periodic access of the blockchain; detection of a virus pattern in the blockchain; detection of an intentionally malformed block in the blockchain; and detection of malicious behavior based on data in the blockchain.
3. The system of claim 1, wherein doing at least one of the following actions includes consolidate the blockchain and bypass the block in the blockchain that comprises the malware and/or anomaly.
4. The system of claim 1, wherein doing at least one of the following actions includes delete the encryption key that was used to encrypt the associated data that comprises the malware and/or the anomaly.
5. The system of claim 1, wherein the microprocessor is further caused to: determine that the malware and/or anomaly is not actually the malware and/or anomaly; and add a block to the blockchain and/or to a consolidation blockchain that indicates that the malware and/or anomaly is not actually the malware and/or anomaly.
6. The system of claim 1, wherein doing at least one of the following actions is based on a type of malware event.
7. The system of claim 1, wherein doing at least one of the following actions is based on a configuration of the blockchain.
8. The system of claim 1, wherein doing at least one of the following actions includes consolidate the blockchain by creating a consolidation genesis block of a consolidation blockchain, wherein the consolidation genesis block identifies the block in the blockchain that has the associated data that comprises the malware and/or the anomaly, and wherein the consolidation genesis block of the

consolidation blockchain points to the blockchain and comprises a hash of the blockchain.

9. The system of claim 8, wherein an end of block is added to the blockchain that points to the consolidation blockchain.

10. The system of claim 9, wherein the end of block identifies the block in the blockchain that has the associated data that comprises the malware and/or the anomaly.

11. A method, comprising: identifying, by a microprocessor, a malware event, wherein the malware event identifies a block in a blockchain that has associated data that comprises malware and/or an anomaly; and in response to identifying the malware event that identifies that the block in the blockchain that has the associated data that comprises the malware and/or the anomaly, doing at least one of the following actions: consolidating, by the microprocessor, the blockchain; consolidating, by the microprocessor, the blockchain and bypassing the block in the blockchain that comprises the malware and/or anomaly; and deleting, by the microprocessor, an encryption key that was used to encrypt the associated data that comprises the malware and/or the anomaly.

12. The method of claim 11, wherein the malware event is based on at least one of the following: detection of a virus in the blockchain; detection of anomalous data in the blockchain; detection of a malicious application/user accessing one or more blocks in the blockchain; detection of abnormal periodic access of the blockchain; detection of a virus pattern in the blockchain; detection of an intentionally malformed block in the blockchain; and detection of malicious behavior based on data in the blockchain.

13. The method of claim 11, wherein doing at least one of the following actions includes consolidate the blockchain and bypass the block in the blockchain that comprises the malware and/or anomaly.

14. The method of claim 11, wherein doing at least one of the following actions includes delete the encryption key that was used to encrypt the associated data that comprises the malware and/or the anomaly.

15. The method of claim 11, further comprising: determining, by the microprocessor, that the malware and/or anomaly is not actually the malware and/or anomaly; and adding, by the microprocessor, a block to the blockchain and/or to a consolidation blockchain that indicates that the malware and/or anomaly is not actually the malware and/or anomaly.

16. The method of claim 11, wherein doing at least one of the following actions is based on a type of malware event.

17. The method of claim 11, wherein doing at least one of the following actions is based on a configuration of the blockchain.

18. The method of claim 11, wherein doing at least one of the following actions includes consolidate the blockchain by creating a consolidation genesis block of a consolidation blockchain, wherein the consolidation genesis blocks identifies the block in the blockchain that has the associated data that comprises the malware and/or the anomaly, and wherein the consolidation genesis block of the consolidation blockchain points to the blockchain and comprises a hash of the blockchain.

19. The method of claim 18, wherein an end of block is added to the blockchain that points to the consolidation blockchain.

20. The method of claim 19, wherein the end of block identifies the block in the blockchain that has the associated data that comprises the malware and/or the anomaly.
